

Comment to the Department of Networking and Information Technology Research and Development (NITRD) and the National Coordination Office (NCO), on behalf of the Office of Science and Technology Policy (OSTP) on the Development of an Artificial Intelligence (AI) Action Plan.

March 15, 2025

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

This submission is not on behalf of the University of California, Berkeley. The views expressed here are the authors' own and do not reflect the views of the University of California, Berkeley.

To the Office of Science and Technology Policy (OSTP),

Thank you for the opportunity to submit comments on the Development of an Artificial Intelligence (AI) Action Plan. My colleagues and I are researchers with expertise in AI research and development, security, and robustness. We offer the following submission for your consideration.

1. Overarching Recommendations

In this section, we provide overarching recommendations on developing an AI Action plan, with an emphasis on risks to national security.

1.1 Prioritize Excellence and Robustness as Key Components of the American Competitive Advantage.

The success/ business value of AI systems is heavily reliant on their security and reliability.¹ Businesses adopting AI products want to avoid unnecessary losses from vulnerable systems, prompting developers to prioritize security alongside innovations.^{2, 3} Additionally, generative

¹ NIST (2023) NIST AI 100-1: Artificial Intelligence Risk Management Framework (AI RMF 1.0). National Institute of Standards and Technology, <https://nvlpubs.nist.gov/nistpubs/ai/nist.ai.100-1.pdf>

² Numerous methods currently exist to jailbreak systems, many of which are not known in advance. See, Abhinav Rao et al. (2024) Tricking LLMs into Disobedience: Formalizing, Analyzing, and Detecting Jailbreaks. arXiv, <https://arxiv.org/pdf/2305.14965>

³ Additionally, models can fail due to hallucinations, data leakage, or prompt injection..

AI models also fail in ways unseen in traditional software,⁴ and some AI applications adapt and change over time, complicating security further. These aspects of AI have drawn attention to critical deficiencies in available testing and evaluation methods.⁵

Implementing insecure systems in critical infrastructure sectors or defense systems could lead to unintended consequences and compromise American national security. Widespread or cascading failures also present challenges, such as AI software supply chain attacks.⁶ Foundation model developers are creating products that will require widespread use and deep integration with current technologies in order to be transformational.⁷

To be an innovation leader, the U.S. must develop state of the art AI systems that are robust, reliable, and secure. This will require new investments in AI security as well as testing and evaluation regimes.

1.2 Maintaining U.S. Leadership in AI

Maintain the work and expertise provided by the U.S. AISI.

The U.S. AISI and its network of AI experts can provide the strategic advantage through government and industry collaboration to operationalize agile governance with the necessary safety nets to accelerate trustworthy⁸ innovation that taps into the potential of AI. It also provides an essential platform for global AI engagement through the broader network of AISIs. To maintain American leadership in paving the way for innovation-centric governance, we recommend increasing investment in testing and evaluation, dissemination of AI best practices and standards, and support of domestic and international institutions advancing AI security and robustness.

⁴ Zhenhua Wang et al. (2024) Foot In The Door: Understanding Large Language Model Jailbreaking via Cognitive Psychology. arXiv, <https://arxiv.org/pdf/2402.15690>

⁵ There are now various platforms and tools available for model and system red teaming, guidelines for secure development across the development and deployment lifecycle, and a strong emphasis on integrating security early on. See, CISA (2024) Joint Guidance on Deploying AI Systems Securely. U.S. Cybersecurity and Infrastructure Security Agency, <https://www.cisa.gov/news-events/alerts/2024/04/15/joint-guidance-deploying-ai-systems-securely>

⁶ Avi Lumelsky et al. (2024) ShadowRay: First Known Attack Campaign Targeting AI Workloads Actively Exploited In The Wild. oligo, <https://www.oligo.security/blog/shadowray-attack-ai-workloads-actively-exploited-in-the-wild>

⁷ Recent security failures across popular products have demonstrated that such systems have become single points of cascading failure. See, GAO (2024a) CrowdStrike Chaos Highlights Key Cyber Vulnerabilities with Software Updates. U.S. Government Accountability Office, <https://www.gao.gov/blog/crowdstrike-chaos-highlights-key-cyber-vulnerabilities-software-updates>

⁸ NIST (2025) Strategic Vision. National Institute of Standards and Technology, <https://www.nist.gov/aisi/strategic-vision>

Champion AI standards and regulations that address national security risks.

The U.S. plays a critical role in establishing and promoting scientifically grounded AI standards that foster innovation while implementing safeguards for high-risk AI applications, as set out during the first Trump administration.⁹ Setting intolerable risk thresholds (supported by 27 countries and the EU at the AI Seoul Summit¹⁰, and the NIST AI RMF¹¹) is one way to mitigate catastrophic risks from AI without slowing down innovation.¹²

Prioritize strategic partnerships that will support international American AI leadership.

Given AI's transnational nature, forging strategic partnerships with allies, leading research institutions, and global organizations is crucial to bolstering U.S. competitiveness and ensuring global AI developments align with American interests.

2. Specific Recommendations on Policy Topics

In this section, we provide recommendations specific to the topics mentioned in the request for information on developing an AI Action Plan.

2.1 Model Development and Deployment

Encourage Phased AI Model Releases

Phased releases, where models and/or model capabilities are deployed incrementally, are a critical component of model security, particularly when dealing with open-weight or open-

⁹ White House (2019) Executive Order on Maintaining American Leadership in Artificial Intelligence. White House, <https://trumpwhitehouse.archives.gov/presidential-actions/executive-order-maintaining-american-leadership-artificial-intelligence/>

¹⁰ DIST (2024) Frontier AI Safety Commitments, AI Seoul Summit 2024. UK DSIT, <https://www.gov.uk/government/publications/frontier-ai-safety-commitments-ai-seoul-summit-2024/frontier-ai-safety-commitments-ai-seoul-summit-2024>

¹¹ "In cases where an AI system presents unacceptable negative risk levels – such as where significant negative impacts are imminent, severe harms are actually occurring, or catastrophic risks are present – development and deployment should cease in a safe manner until risks can be sufficiently managed." in page 8 of NIST (2023) NIST AI 100-1: Artificial Intelligence Risk Management Framework (AI RMF 1.0). National Institute of Standards and Technology, <https://nvlpubs.nist.gov/nistpubs/ai/nist.ai.100-1.pdf>

¹² Deepika Raman et al. (2025) Intolerable Risk Threshold Recommendations for Artificial Intelligence. UC Berkeley Center for Long-term Cybersecurity, <https://cltc.berkeley.edu/wp-content/uploads/2025/02/Intolerable-Risk-Threshold-Recommendations-for-Artificial-Intelligence.pdf>

source models.¹³ This is critical in the context of open-weight models, which have a limited number of post-deployment mitigations,¹⁴ and high-risk domains.

- Deploying AI models in phases gives researchers and security experts the ability to identify and mitigate vulnerabilities before they are abused by malicious actors.
- Phased releases also ensure that AI models comply with security regulations and standards, particularly as these frameworks rapidly evolve.

2.2 Open Source Development and Robustness

Implement AI security regulations alongside guidance to ensure best practices are appropriately applied to both closed and open-source models. Fostering an open-source community focused on creating secure and robust models will require investments in shared computing and data resources, further enabling reliable innovation. These communities will also allow model developers to share critical security insights and work together to tackle the nation's most threatening AI risks.

The open-source approach to security in traditional software development, which relies on a great number of experts discovering and remediating vulnerabilities, has proven to be a successful way to make systems secure.¹⁵ However, even with traditional software, there are tradeoffs and use cases where closed-source alternatives provide more security.¹⁶ While many of the core learnings from open-source security apply to AI, the unique nature of AI models necessitates a tailored approach to risk management. See section 2.1 for some recommended approaches.

¹³ See, for example, the [OpenAI \(2023\)](https://cdn.openai.com/openai-preparedness-framework-beta.pdf) preparedness framework (<https://cdn.openai.com/openai-preparedness-framework-beta.pdf>), and the [Anthropic \(2023\)](https://www-cdn.anthropic.com/1adf000c8f675958c2ee23805d91aaade1cd4613/responsible-scaling-policy.pdf) responsible scaling policy (<https://www-cdn.anthropic.com/1adf000c8f675958c2ee23805d91aaade1cd4613/responsible-scaling-policy.pdf>), which both include requirements for phased releases and/or structured access, with efforts to detect and respond to misuse or problematic anomalies.

¹⁴ Madhulika Srikumar et al. (2024) Risk Mitigation Strategies for the Open Foundation Model Value Chain. Partnership on AI, <https://partnershiponai.org/resource/risk-mitigation-strategies-for-the-open-foundation-model-value-chain/#:~:text=Component%2Dlevel%20releases,of%20openness%20while%20mitigating%20risks>

¹⁵ Russell Clarke et al. (2009) Is Open Source Software More Secure? University of Washington, [https://courses.cs.washington.edu/courses/csep590/05au/whitepaper_turnin/oss\(10\).pdf](https://courses.cs.washington.edu/courses/csep590/05au/whitepaper_turnin/oss(10).pdf)

¹⁶ Closed systems allow for a higher degree of control over the development and patching process, which can be beneficial when strict quality control and consistency are key. This also allows for tighter control over distribution, which can be crucial for AI systems that possess specialized capabilities.

2.3 Explainability and Assurance of AI Model Outputs

Require that AI companies openly share the results of their testing and evaluation with governments and independent researchers, enabling oversight and accountability. Without such transparency, there is a significant risk that companies could downplay or conceal safety concerns.¹⁷ Additionally, transparency in model evaluations, beyond just evaluation results, provide visibility into their robustness and accuracy. Especially in the context of increasingly capable AI systems deployed in critical domains, transparency from leading AI labs can go beyond ensuring secure development to even mitigating national security risks. Appropriate transparency from model developers throughout the AI lifecycle can support quicker consensus on best practices.

2.4 Regulation, Government, and Standards

High-priority risks, such as illegal imagery (e.g., CSAM), CBRN weapons, cybersecurity risks, and AI autonomy, require strong federal oversight. **AI regulation that addresses national security risks must be proactive and enforceable, with the U.S. leading the way in setting clear, effective policies that serve as a global standard.** Leaving regulation to individual states may create a tapestry of guidelines that may be conflicting or inconsistent, placing more regulatory burden on model developers and increasing the risk of legal uncertainty leading to unaddressed catastrophic risks. Such proactive governance will be most efficiently enacted only through sufficient international cooperation that deters any escalation of such risks.

2.5 Cybersecurity, Data Privacy and Security Throughout the Lifecycle of AI system Development and Deployment (to include security against AI model attacks)

Incentivize Developing Strong Mitigation for Evaluation Deception Risks.

The U.S. government needs to incentivize the development of strong mitigations for evaluation deception by AI models before they possess this capability at an unacceptable level that threatens U.S. national security. Evaluation deception (i.e. a model being capable of substantially and persistently deceiving its evaluators about its capabilities) is an emerging security risk. A model successfully deceiving its evaluators would not only heighten loss-of-

¹⁷ This may lead to harmful consequences, similar to the tobacco industry's history of strategically undermining emerging science on health risks. See, Allan M Brandt (2012) Inventing Conflicts of Interest: A History of Tobacco Industry Tactics. National Library of Medicine, <https://pmc.ncbi.nlm.nih.gov/articles/PMC3490543/>

control risks, but also increases the likelihood of dangerous levels of model capabilities going unidentified.¹⁸

Safeguard the Digital-to-Physical Frontier.

It is important to invest in research that addresses and evaluates national security threats posed by systems at the digital-to-physical frontier (e.g., AI controlled infrastructure and robotics). Threats from AI-driven physical systems pose a unique challenge for security and have the potential to cause large scale harm.¹⁹ As AI capabilities extend beyond digital spaces, risks to US national security may escalate, introducing unanticipated threats that we may not yet fully understand, and that current safeguards are ill-equipped to handle. Without proactive investment in research, the U.S. risks falling behind adversarial actors who may weaponize these systems in ways that threaten national security.

Protect against model theft.

Long-term protection against the most advanced threats to model weights protection demands sustained investment and research prioritization. It is critical that the necessary controls and standards are created and enforced for developers of AI models with dangerous capabilities.²⁰ Adversarial actors gaining access to model weights may lead to leveraging advanced AI capabilities as weapons (e.g., cyber-attacks, and autonomous threats).

2.6 National Security and Defense

Utilize the talent and expertise based within NIST and the U.S. AISI by tasking these institutes with tackling the most critical national security capabilities and risks. This includes ongoing support and expansion of existing initiatives aimed at addressing the most critical risks (listed below) to U.S. national security. These efforts include, but are not limited to:

- NIST AI 100-1: [AI Risk Management Framework](https://nvlpubs.nist.gov/nistpubs/ai/nist.ai.100-1.pdf) (RMF) ²¹

¹⁸ Deepika Raman et al. (2025) Intolerable Risk Threshold Recommendations for Artificial Intelligence. UC Berkeley Center for Long-term Cybersecurity, <https://cltc.berkeley.edu/wp-content/uploads/2025/02/Intolerable-Risk-Threshold-Recommendations-for-Artificial-Intelligence.pdf>

¹⁹ RAND (2025) Mitigating Risks at the Intersection of Artificial Intelligence and Chemical and Biological Weapons. RAND, https://www.rand.org/pubs/research_reports/RRA2990-1.html

²⁰ RAND (2024) Securing AI Model Weights. RAND, https://www.rand.org/pubs/research_reports/RRA2849-1.html

²¹ NIST (2023) NIST AI 100-1: Artificial Intelligence Risk Management Framework (AI RMF 1.0). National Institute of Standards and Technology, <https://nvlpubs.nist.gov/nistpubs/ai/nist.ai.100-1.pdf>

- NIST AI 600-1: [Generative AI Profile](#) ²²
- The U.S. AI Safety Institute Consortium (AISIC) working groups and task forces
- NIST SP 800-218A: [Secure Software Development Practices for Generative AI and Dual-Use Foundation Models: An SSDF Community Profile](#) ²³

2.7 International Collaboration

While U.S. AI dominance requires robust domestic capabilities, **international collaboration is critical for sustained U.S. AI leadership and competitiveness.** Given AI's transnational nature, forging strategic partnerships with allies, leading research institutions, and global organizations is crucial to bolstering U.S. competitiveness and ensuring global AI developments align with American interests. Effective communication, coordination, and collaboration between the U.S. government and international stakeholders will be essential in establishing formal AI regulations as these emerging technologies continue to evolve and introduce increasingly complex regulatory challenges.²⁴

2.8 Research and Development

Investing in AI research and education is critical for maintaining U.S. leadership in the global AI race. Advancing cutting-edge research fosters innovation, ensures technological superiority, and strengthens national security by keeping American AI systems at the forefront of development. It is also equally important to retain existing talent and further support U.S. institutes (e.g., NIST and U.S. AISI) that harbor expertise that is critical to U.S. global AI leadership.

A strong AI education pipeline equips the workforce with the necessary skills to drive progress, attract top global talent, and sustain a competitive edge in AI-driven industries. Without a well-funded and structured approach to AI research and education, the U.S. risks falling behind countries that prioritize these areas, losing its influence over global AI standards, and diminishing its technological and economic leadership.

²² NIST (2024) NIST AI 600-1: Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile. National Institute of Standards and Technology, <https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf>

²³ NIST (2024) NIST SP 800-218A: Secure Software Development Practices for Generative AI and Dual-Use Foundation Models: An SSDF Community Profile. National Institute of Standards and Technology, <https://csrc.nist.gov/pubs/sp/800/218/a/final>

²⁴ RAND (2025) Mitigating Risks at the Intersection of Artificial Intelligence and Chemical and Biological Weapons. RAND, https://www.rand.org/pubs/research_reports/RR2990-1.html

Our best,

Nada Madkour, Ph.D.

Non-Resident Research Fellow

AI Security Initiative, Center for Long-Term Cybersecurity, UC Berkeley

Krystal Jackson

Non-Resident Research Fellow

AI Security Initiative, Center for Long-Term Cybersecurity, UC Berkeley

Deepika Raman

Non-Resident Research Fellow

AI Security Initiative, Center for Long-Term Cybersecurity, UC Berkeley

Evan R Murphy

Non-Resident Research Fellow

AI Security Initiative, Center for Long-Term Cybersecurity, UC Berkeley

Jessica Newman

Director

AI Security Initiative, Center for Long-Term Cybersecurity, UC Berkeley

Co-Director

AI Policy Hub, UC Berkeley